

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Supported the review of **system requirements** and architecture designs to assess testability and plan test coverage, collaborating with System Engineers and Function Owners on enterprise applications.
- Designed and implemented test specifications fulfilling **system functional and performance requirements** for automotive camera systems and mechatronic ECU systems, achieving **95% requirement coverage**.
- Executed system testing aligned with project schedules, including **stress and robustness testing** of system features, monitoring test progression and reporting deviations.
- Developed automated test scripts using **Python**, adhering to platform standards and project automation targets, improving test execution efficiency by **30%**.
- Performed **trace analysis and troubleshooting** to determine root cause of issues and validate fixes for system-level defects, reducing bug resolution time by **15%**.
- Maintained comprehensive test results, test logs, and test reports throughout the project lifecycle, ensuring **traceability of system test specification** to system requirements.

SKILLS

Programming Languages – Python, C++, Java

Testing Frameworks & Tools – NI LabVIEW, NI TestStand, Jenkins, Git

Systems Engineering – Systems Development Life Cycle, System Requirements, Architecture Design, Test Specification, System Validation, Traceability

Automotive Systems – ADAS, Automotive Camera Systems, Mechatronic ECU Systems, Bench Testing, HIL Testing, Vehicle Testing

Development Tools – Jira, Confluence, Docker

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Automotive Sensor Fusion Test Harness

- Developed a **Python-based test harness** to simulate sensor inputs for ADAS systems, integrating with existing **bench testing** environments.
- Designed test cases for validating sensor fusion algorithms against **performance requirements**, covering edge cases and failure modes.

- Implemented automated data logging and reporting mechanisms, providing detailed **test logs** for root cause analysis of identified defects.

System Validation Framework for Embedded Automotive Software

- Contributed to the design of test specifications for **embedded automotive software features**, ensuring alignment with system-level requirements and standards.
- Implemented test automation scripts using **Python and Bash** to execute test plans on target hardware, achieving **85% test automation coverage** for core functionalities.
- Collaborated with System Engineers to perform **trace analysis**, mapping test results back to system requirements and documenting validation status.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)